# ELEC4605 ASSIGNMENT 1 – TERM 3, 2019

## NOISY QUANTUM SYSTEMS

In this assignment, you will be applying the filter function formalism presented in the lectures, along with a density matrix and rotation operator analysis to study the effect of noise on quantum systems. As in the lectures, we will consider spin as our prototypical quantum system. It is advised you review the section on "controlling quantum systems" in your lecture notes before attempting this assignment.

You are to investigate an ensemble of spins in an environment where they are subject to noise from different sources. You can assume that the noise is stochastic (a random process) and both stationary and ergodic. Whilst the origin of the noise is uncertain, you can assume that it couples to the spin magnetically (i.e. it appears as an effective magnetic field noise). The noise causes dephasing of the spins, resulting in a decay of their average coherence over time. We have recorded some sample traces of the effective magnetic field noise, which are provided to you in two different formats: the first as a matrix in a MAT-file for importing into MATLAB (should you choose to use this for your analysis), and the second as a comma separated variable file (for importing into other numerical software packages). Both files contain the same noise samples. You may access this data through the following link: https://www.dropbox.com/sh/4gvvp71itw2g47u/AABIAt6P6rKnl0IZu4dVbF-1a?dl=0.
The traces were sampled with a time resolution of 2.5 μs and are in units of magnetic field (Tesla). The individual time traces run along each row, with 100 sample traces provided.

**(a)** Use the filter-function theory to plot the decay of coherence of the spin ensemble due to this noise for the following experimental pulse sequences:
(i) Ramsey (free induction decay) [*10 marks*]
(ii) Hahn echo [*10 marks*]
(iii) A multi-pulse dynamical decoupling sequence with increasing number of pulses and a fixed duration between them. It's up to you to select an appropriate pulse separation time (or perhaps you might wish to consider a few different cases). [*10 marks*]

In each part above, you should plot the decay of the coherence as a function of the total pulse sequence length ($\tau$) and fit each curve <u>with an appropriate function</u> to extract the coherence/dephasing time. Note, you will need to calculate the noise power spectral density to complete this task, you may either do this directly by taking the Fourier transform of the autocorrelation function of the noise, or you can use a built-in function from your numerical software package (e.g. `pwelch();` in MATLAB). Be sure to include a discussion of your results.

**(b)** Use the same noise signals to perform a "brute-force" simulation of the decoherence by calculating density matrices and applying rotation operators. Here each noise trace can be viewed as the noise experienced by individual spins in the ensemble. You can assume that all pulses are instantaneous.
(i) Simulate the decay of the coherence for each of the pulse sequences listed above (plot the decay curves and extract coherence times) and compare the results with part a [*25 marks*]
(ii) List the pro's and con's of the filter function and brute-force methods [*10 marks*]

**(c)** In this part you are to use the spins to probe the spectral density of the noise:
(i) Use the brute-force multi-pulse dynamical decoupling simulations to reconstruct the power spectral density (PSD) of the noise. You can only use information extracted from the coherence decay measurements in conjunction with your knowledge of the pulse sequence applied. Be as quantitative in extracting the noise PSD as possible and justify your choice of pulse parameters (pulse time separation, number of pulses etc). [*25 marks*]
(ii) Discuss the PSD you have extracted and comment on the possible origins of the noise. [*10 marks*]

**Submission:** You are expected to complete this assignment and write a report summarising your results by yourself. The report is due on Friday, week 5 (18/10/2019) at 6pm. Submission will be via Turnitin on the Moodle course page. This assignment is worth 12.5% of your total grade for the course and will be marked out of 100.